Enterprise / Small Business (SOHO) Network

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1.Project Objective -

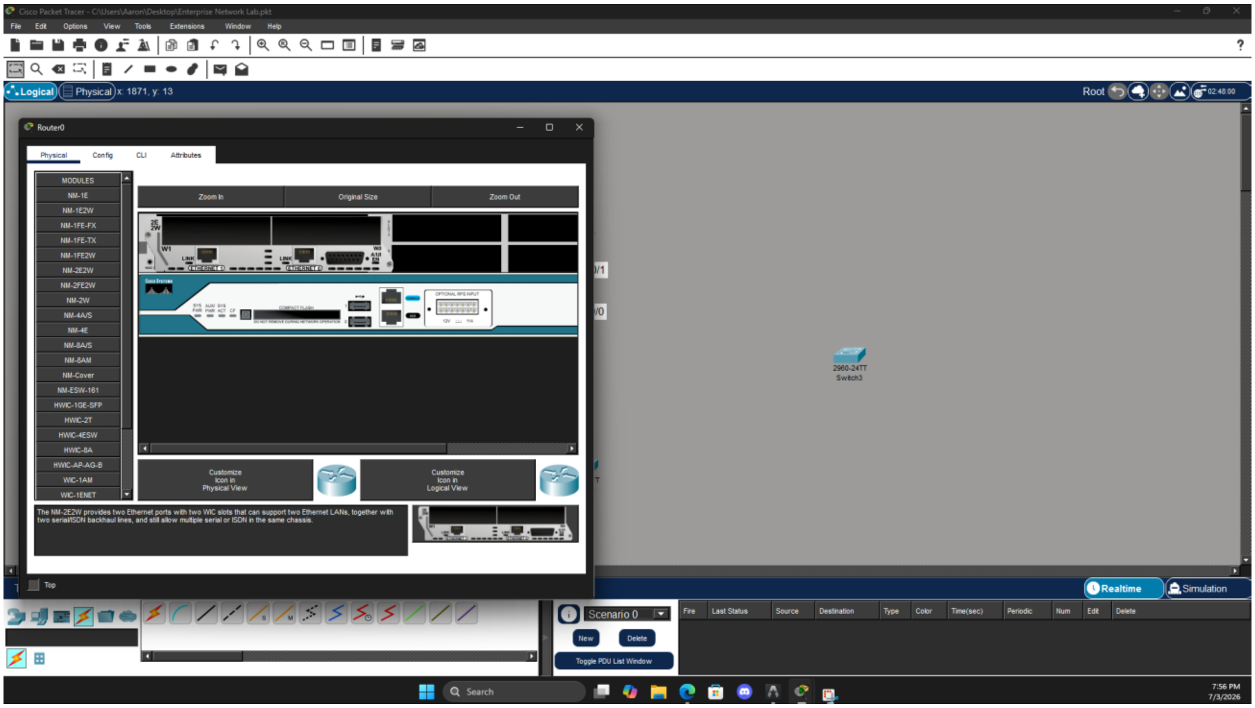
The goal of this project was to design, build, and test a functional small business SOHO; along with producing errors, that would be noticed, deduced, and fixed using logical reasoning in foundational and basic IT knowledge. The main application used to create a logical and physical artifact was Cisco Packet Tracer; I created a multi-switch topology with a central router, a server, and several end-client PCs. To make this exercise more realistic and educational, I intentionally introduced a configuration error, documented the troubleshooting process, fixed the issue, and verified full connectivity across the network. I then induced a basic knowledge issue, which is an issue that most never be done in an enterprise network, due to problematic configurations and VLAN concepts and managements, that will go unknown due to local connectivity still being possible.

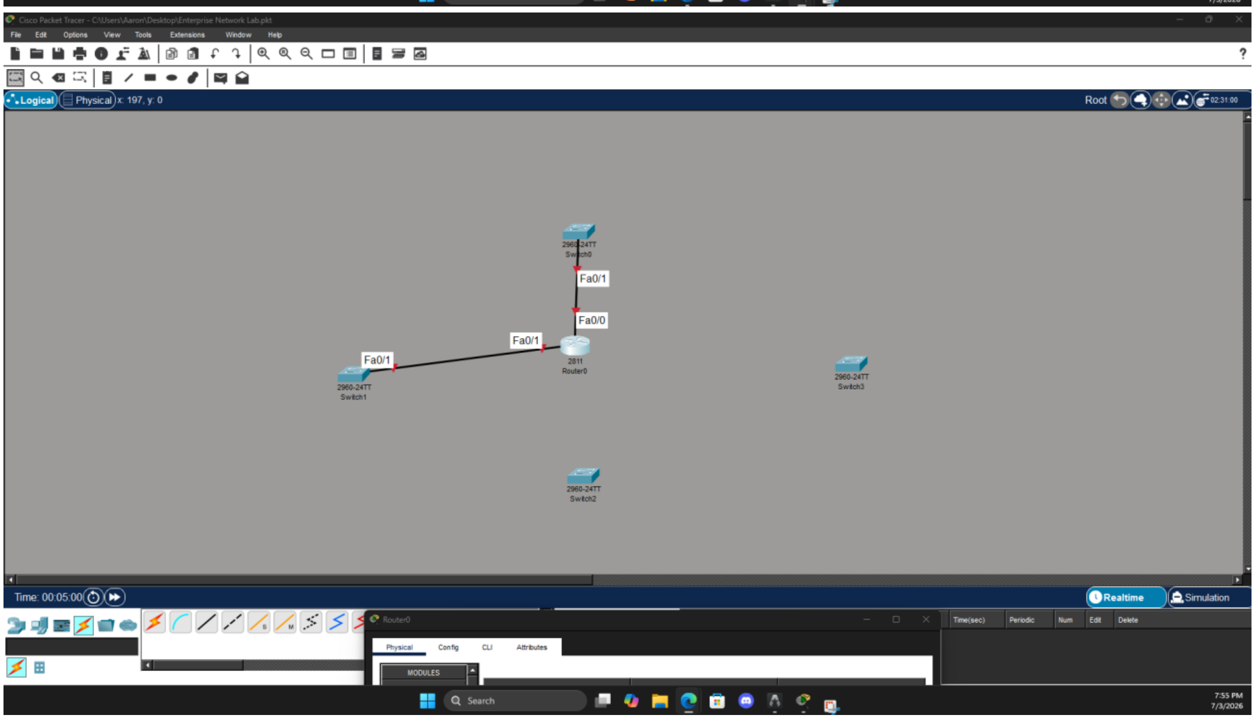
2. Physical Topology -

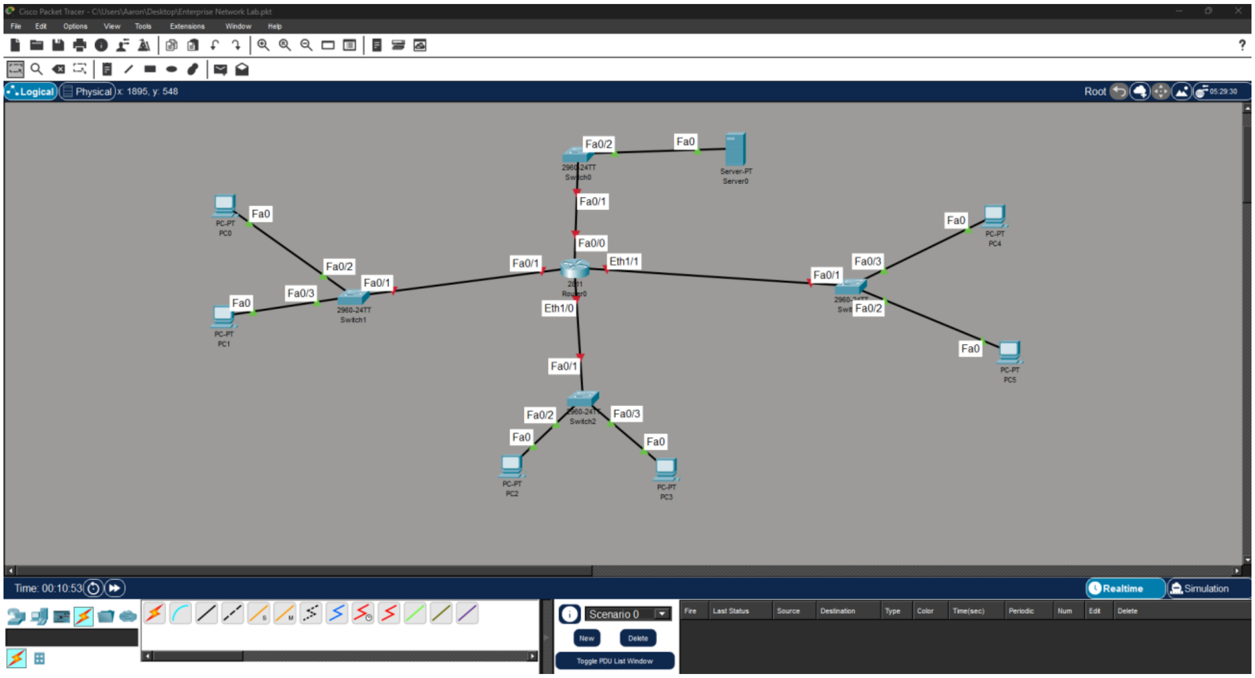
I placed the following devices and connected them using straight-through Ethernet cables:

- 1 Router (R1).
- 4 Switches (SW1, SW2, SW3, SW4).
- 1 Server.
- 6 PCs.

To support additional connections, I added a NM-2FE2W module to the router for extra Ethernet ports. The devices were then wired together, using straight-through Ethernet (note: real world will have structured cabling, and rooms designated for IT equipment).


Router0
Physical
Config
CLI
Attributes
MODULES
Zoom In
Original Size
Zoom Out
Customize Icon in Physical View
Customize Icon in Logical View
The NM-2E2W provides two Ethernet ports with two WIC slots that can support two Ethernet LANs, together with two serial/ISDN backhaul lines, and still allow multiple serial or ISDN in the same chassis.
2960-24TT
Switch3
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window


2960-24TT
Switch0
Fa0/1
Fa0/0
Fa0/1
Fa0/1
2811
Router0
2960-24TT
Switch1
2960-24TT
Switch3
2960-24TT
Switch2
Realtime
Simulation



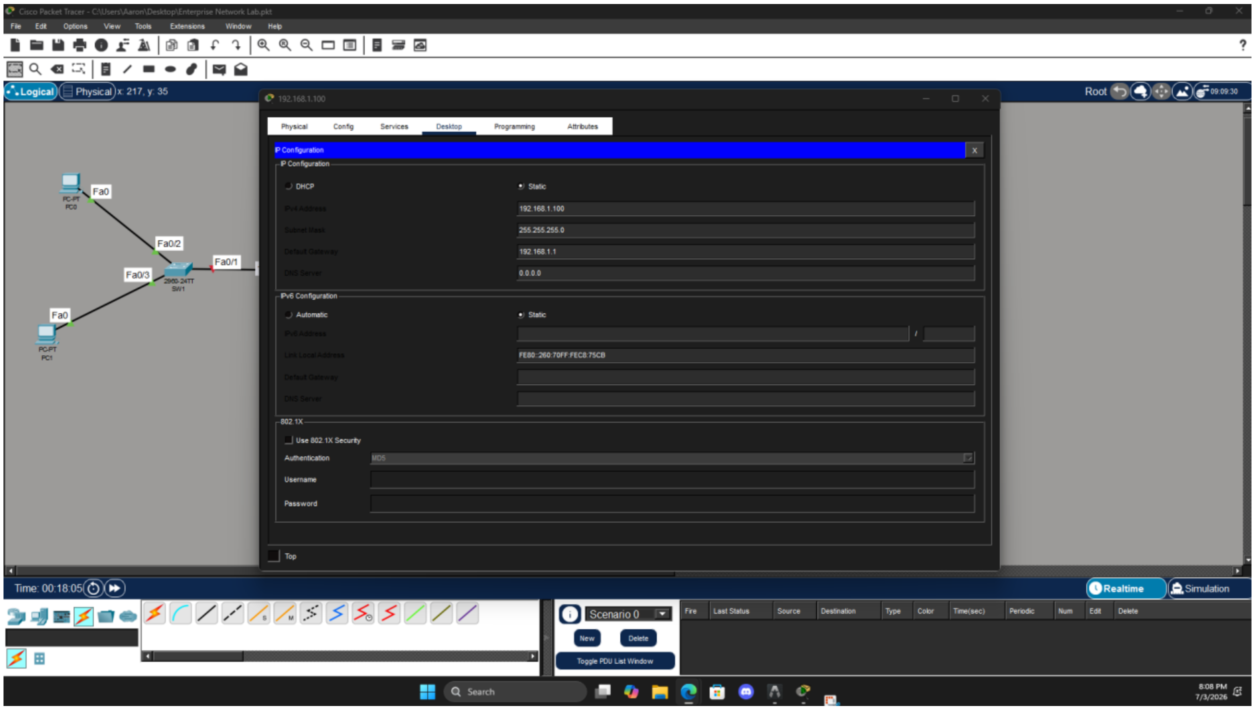
The connections are created, using the router as the star, that connects four switches to prevent clustering, and creates secure segments.

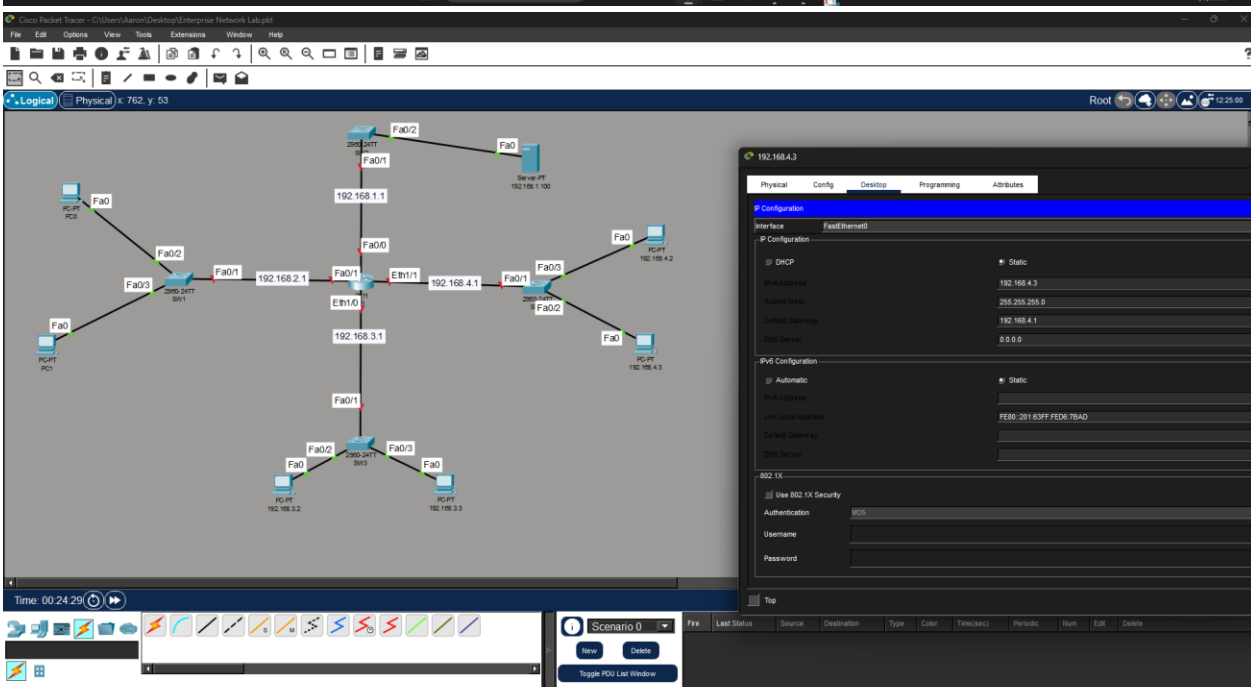
3. Logical Topology and IP Addressing -

A simple scheme was designed for this network:

- 192.168.1.0/24 - Management/Server (SW2).
- 192.168.2.0/24 - Devices connected to SW1.
- 192.168.3.0/24 - Devices connected to SW3.
- 192.168.4.0/24 - Devices connected to SW4.

The router serves as the default gateway for all segments. (Remember, there is a purposed error within; a duplicated IP that will be critical if the network decides to use VLANs and VLAN management. It still works due to the fact that switches work at layer 2, where you don't need an IP for basic switching and pinging between devices. Subnetting also allows it to work, until you indefinitely need VLANs).

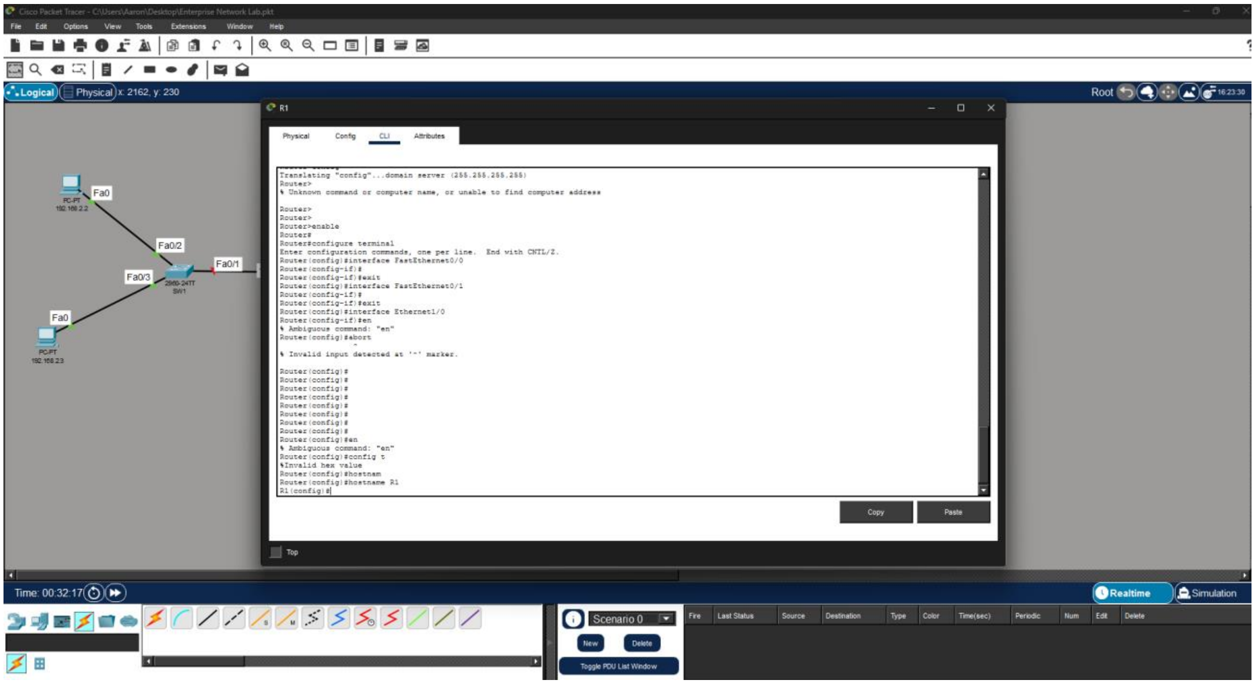


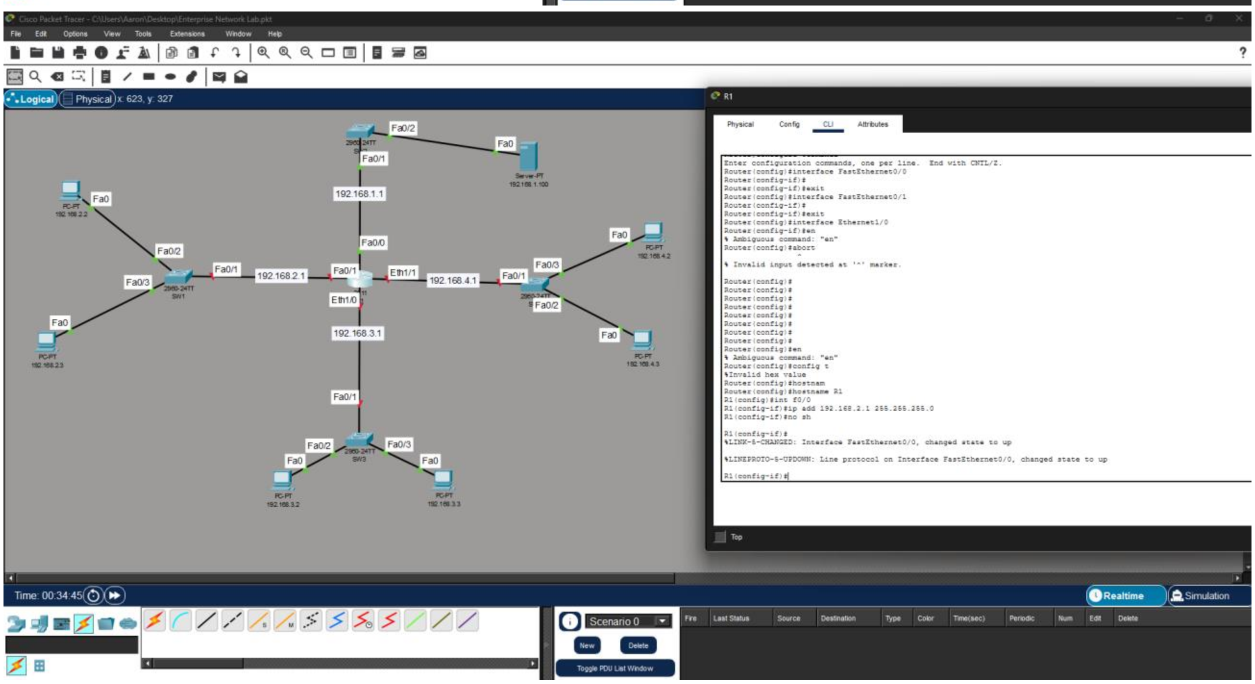


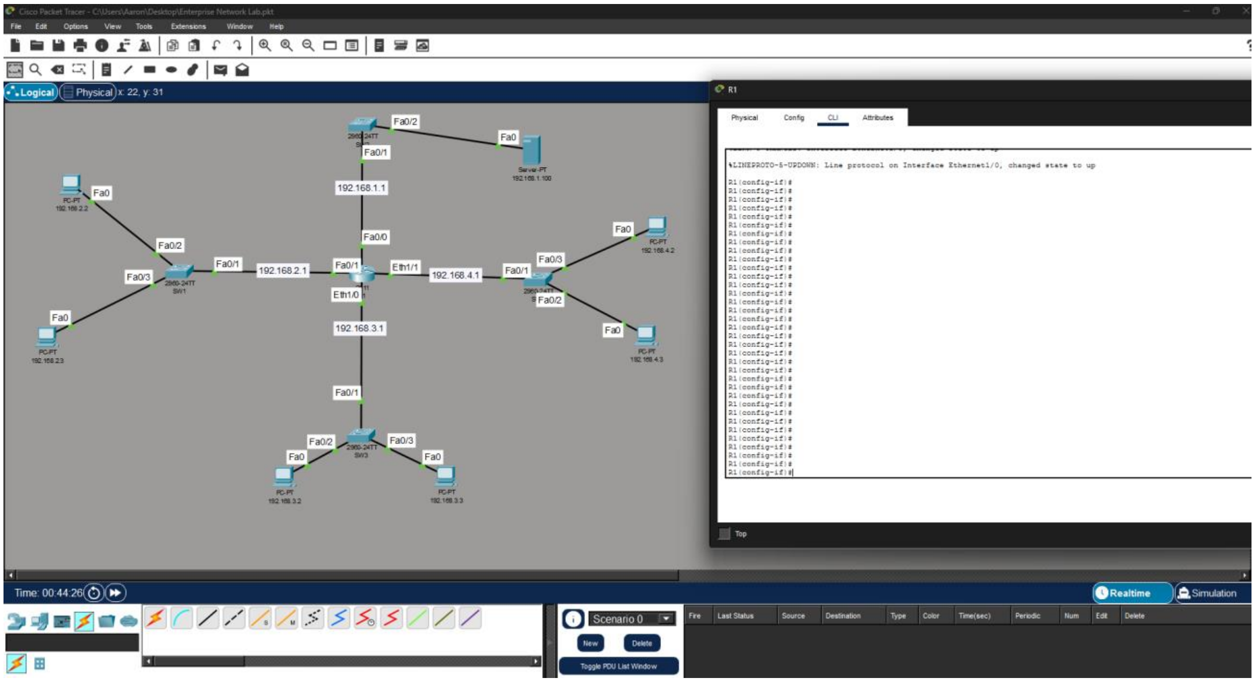
The server uses the router's IP for its default gateway, which is 192.168.1.1 with the server itself being 192.168.1.100. This is crucial to pay attention to, or if you already know, you know that this step is critical for management and proper addressing.

4. Configuration -

I configured the router hostname and interfaces via CLI. Switches were set up for basic operations. Using the proper commands, such as hostname to name the Router to R1, int and path to select said path, and IP add to add the IP and ensure the path state is up. I used a mix of FastEthernet (100 Mbps) and standard Ethernet (10 Mbps) ports. FastEthernet provides significantly higher bandwidth and is better suited for connections between the router and switches, while basic Ethernet is adequate for low-traffic end devices but can become a bottleneck in production environments. Understanding the differences helped guide port assignments, and it is critical to know your physical cabling and ports. It is critical to know your CLI, and it is critical to know your cables and hardware.


R1
Physical
Config
CLI
Attributes
Translating "config"...domain server (255.255.255.255)
Router>
% Unknown command or computer name, or unable to find computer address
Router>
Router>
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Ethernet1/0
Router(config-if)#en
% Ambiguous command: "en"
Router(config)#abort
% Invalid input detected at '^' marker.
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#en
% Ambiguous command: "en"
Router(config)#config t
%Invalid hex value
Router(config)#hostnam
Router(config)#hostname R1
R1(config)#
Copy
Paste
Top
Fa0
Fa0/2
Fa0/1
Fa0/3
Time: 00:32:17
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window

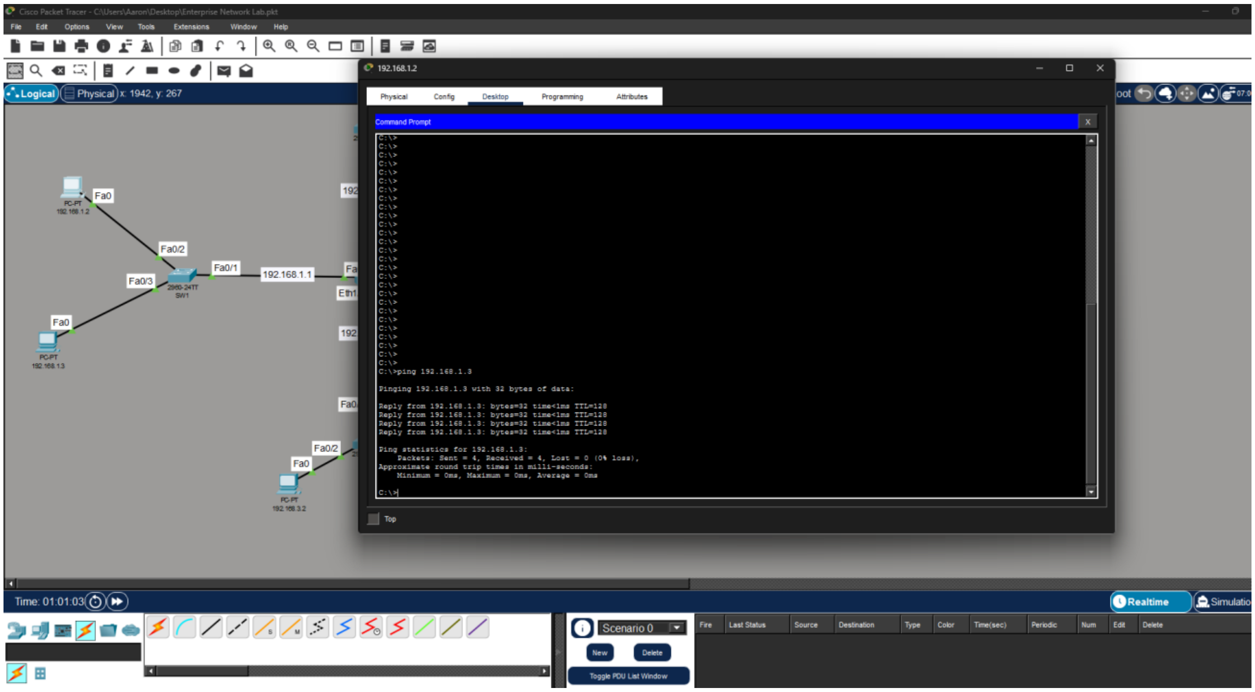

192.168.1.1
192.168.2.1
192.168.4.1
192.168.3.1
Fa0/0
Fa0/1
Eth1/1
Eth1/0
Fa0/2
Fa0/3
Fa0
R1
Physical
Config
CLI
Attributes
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Ethernet1/0
Router(config-if)#en
% Ambiguous command: "en"
Router(config)#abort
% Invalid input detected at '^' marker.
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#en
% Ambiguous command: "en"
Router(config)#config t
%Invalid hex value
Router(config)#hostnam
Router(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip add 192.168.2.1 255.255.255.0
R1(config-if)#no sh
R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#
Top
Time: 00:34:45
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window



5. Testing and Troubleshooting (Initial error).

Initial testing:

Devices on the same switch could ping each other successfully. The state was up for the switch allowing the PCs on the same segment to talk as the switch forwards the information.



The error created:

PCs on different switches and the server could not communicate reliably. Pings resulted in "Request timed out".

Diagnosis:

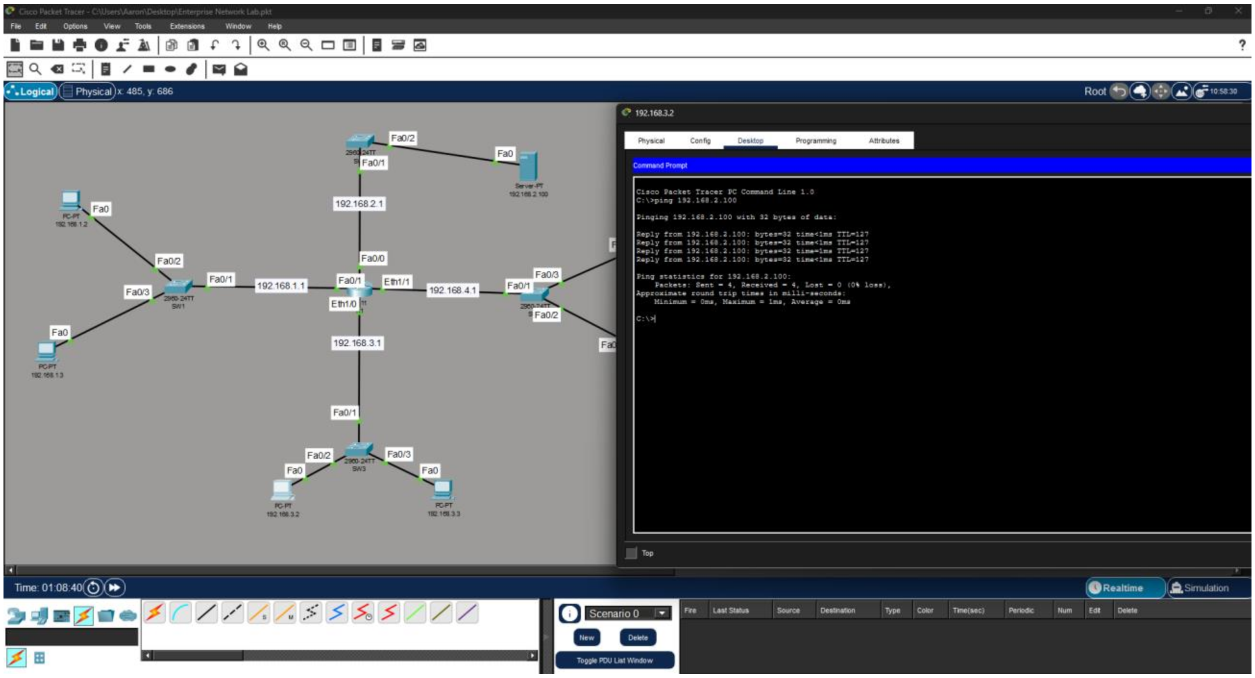
I checked the IP configurations on the PCs and server. The issue was incorrect or missing Default Gateway settings on several PCs – they were not pointing to the router interface on their respective subnets.

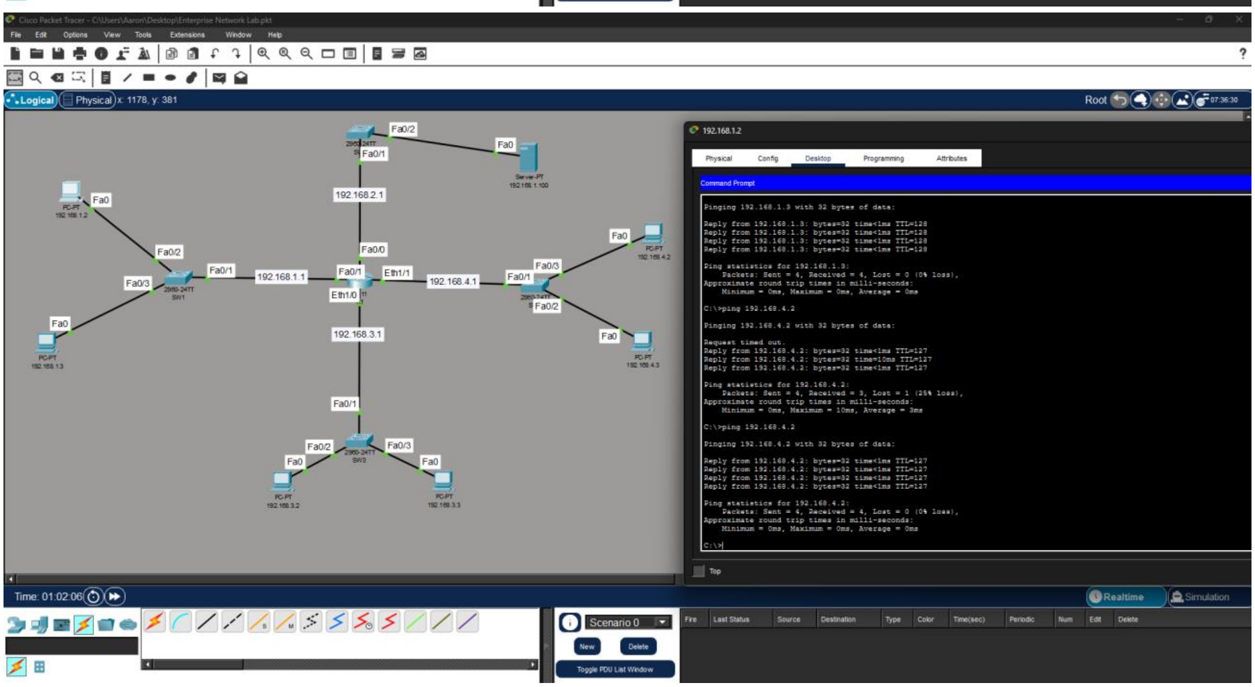
Resolution:

1. Went to each PC, respective desktop, and their IP configuration.
2. Assigned correct static IP, Subnet Mask, and Default Gateway.
3. Verified router interfaces.
4. Re-tested pings.

Result:

Full connectivity achieved, however, there is still one major issue that will prevent reliable connection to the server from the main PCs and for sure, VLAN issues if this was a real environment. It is a golden rule to not duplicate IPs and ensure configuration is properly done with the proper addresses; you may get errors that are "gimmicky" or "finicky" if not.





6. Additional Note on Switch Management IP -

In this lab, the switches were not assigned management IP addresses. This is not necessary for basic layer 2 switching and ping functionality, as the switches successfully forwarded traffic between devices.

In a production environment, assigning a management IP (192.168.1.2 on VLAN 1) to switches is crucial for:

- Remote administration via Telnet or SSH.
- Monitoring and logging.
- Centralized management from a network operation center.

Switches should never share or duplicate another IP or IP of the router, as it is the golden rule. A switch may still work, but it can get confused; packets can drop or go to the wrong location, and pings may or may not work. This is also demonstrated by 192.168.1.1 (you may need to download the file or recreate this using Cisco).

I noted this during review and will include management IPs in future, more advanced labs.

7. Skills Demonstrated -

- Designing physical and logical network topologies.
- Proper cabling and device configuration in Cisco Packet Tracer.
- IP addressing, subnetting, and default gateway setup.
- Systematic troubleshooting connectivity issues.
- Professional documentation of network projects.

8. Conclusion -

This lab successfully created a working small business network. By intentionally introducing and resolving a common configuration error, I gained practical experience that directly applies to entry-level IT and networking roles. The final topology allows reliable communication between all devices. It also expands a "golden rule" that will result in a major issue if the company pursues expanding and using VLANs. Duplicate IPs are a no go and should be avoided. Along with knowing the routers IP, including the default gateway.